

kNN Regression and kNN and Naive Bayes Classification on Baseball Data

Sophia Gavasheli and Jacob Carlson

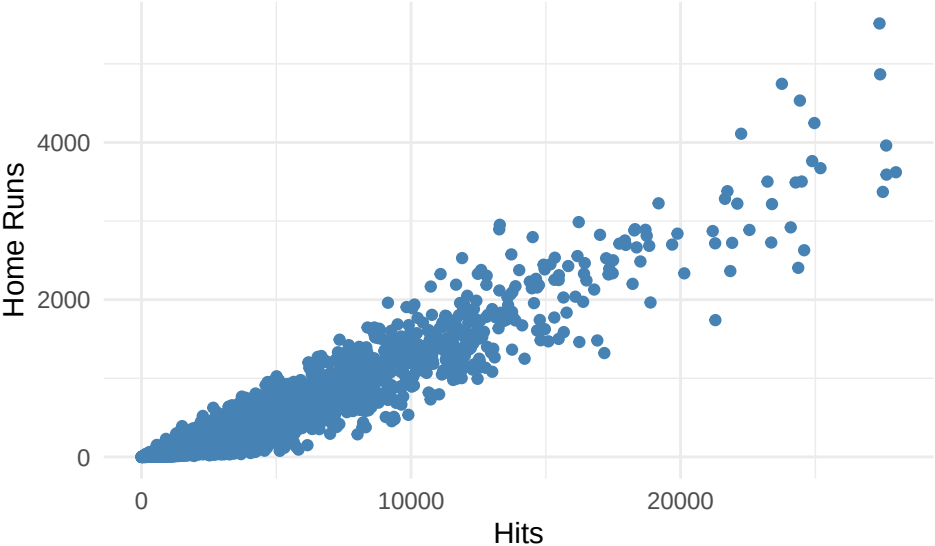
October 8, 2026

Introduction & Research Questions

Exploratory Data Analysis (EDA)

- Bullet 1
- Bullet 2
- Bullet 3

Hits vs Home Runs



Hits Across Teams by Position

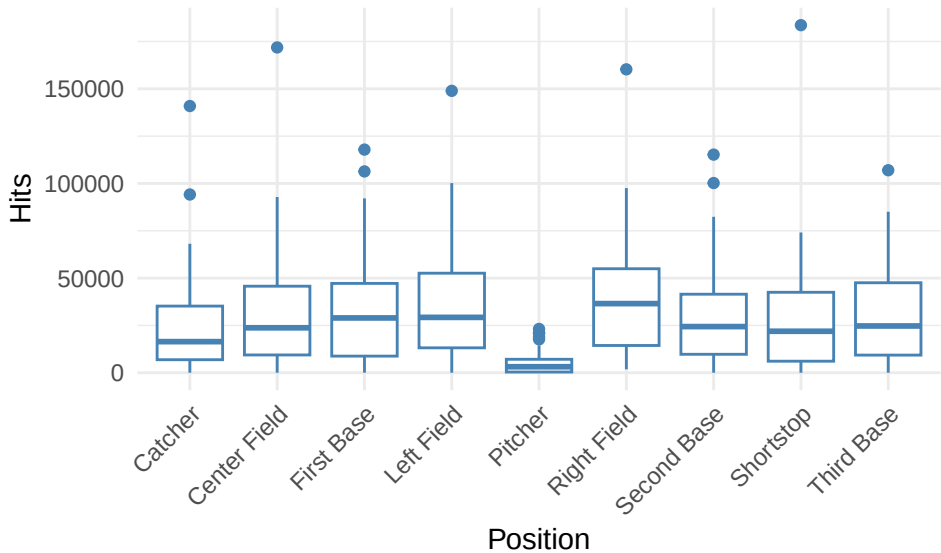


Table 1: Test-set comparison of the no-skill baseline, kNN, and Naive Bayes.

Metric	No-Skill	kNN ($k^* = 51$)	Naive Bayes
Log-Loss	1.108	1.064	1.069
Brier Score	0.633	0.618	0.622
Balanced Accuracy (Macro)	0.500	0.556	0.563
Macro-F1	0.369	0.345	0.336

Calibration Plot

